

FutureXcel (Week # 1)

Data Science

Official Task Instructions:

Data Science Interns

WEEK 1 — Advanced Data Cleaning & Preprocessing

What it is: Make messy real-world data analysis-ready.

Task: Choose a dataset ($\geq 5,000$ rows) and perform deep cleaning.

Must contain:

- Load dataset & initial diagnostics
- Missing value analysis (visual & table)
- Imputation using ≥ 2 techniques (e.g., mean, KNN)
- Data type corrections & date parsing
- Outlier detection & documented treatment (IQR or z-score)
- Final clean dataset (CSV)

Hard details:

- Log all transformations in a reproducible notebook/script
- Provide before/after sample rows

Submission: PDF with code screenshots + cleaned CSV + LinkedIn post

